

Course- B. Tech
Course Code- CSET343
Year- 4th Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Lab Assignment - 4

Objective: This assignment aims to provide hands-on experience with logistic regression, a fundamental algorithm for binary classification problems. You will work with a real-world dataset to preprocess data, train and evaluate a logistic regression model, and explore advanced concepts such as regularization and model interpretation

Dataset: [Breast Cancer Wisconsin \(Diagnostic\)](#)

Tasks:

Task 1: Data Acquisition and Exploration

1. Download and load the dataset
2. Compute summary statistics (mean, median, standard deviation, etc.) for all features.
3. Check for missing values, duplicates, or outliers. Handle them
4. Visualize the data: Create histograms or box plots for key features, a correlation heatmap, and class distribution.
5. Discuss any imbalances in the class distribution and potential implications for model training.

Task 2: Data Preprocessing

- Encode the target variable (Outcome) as binary.
- Split the data into training and testing sets (e.g., 80/20 split) using stratified sampling to maintain class balance.
- Standardize the features to ensure they are on the same scale, as logistic regression is sensitive to feature scales.

Task 3: Model Training and Basic Evaluation

- Train a logistic regression model
- Evaluate the model on the test set using the following metrics:
 - Accuracy
 - Precision, Recall, and F1-score (report for both classes)
 - Confusion matrix (visualize it)
 - ROC curve and AUC score

Interpret the results: Discuss what these metrics reveal about the model's performance, especially in the context of a medical diagnosis task.